

Appendix: Transparent AI Usage Report

Course: Expert Systems

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1 Overview and Compliance Statement

In strict compliance with the course policy on Artificial Intelligence usage, this document provides a comprehensive, transparent audit of the AI tools utilized during the research, implementation, and writing phases of the final project, titled “*The Illusion of Inference: A Neuro-Symbolic Falsification of LLM Reasoning via Formal Verification and Token Surprisal*”.

The human author (**Alireza Ebrahimi**) maintains 100% intellectual responsibility and accountability for all theoretical formulations, code execution, dataset verifications, statistical analyses, and final manuscript conclusions. AI assistance was utilized strictly as an interactive research accelerator, code reviewer, and technical editor.

2 AI Tool Specifications

Table 1 details the specific AI systems, versions, and deployment modalities used throughout the project lifecycle.

Table 1: Specification of AI Assistance Tools

Tool / Model	Provider	Primary Purpose in Project
Gemini (Deep Research Mode)	Google	Initial literature exploration and identification of seminal open-access papers (e.g., FoVer, Robust Analogies).
Gemini 3.1 Pro / 3.6 Flash	Google	Scaffolding initial Python scripts (spaCy NER, PyTorch logit extraction, Z3 SMT logic mapping) and LaTeX formatting.

3 Key Prompts and Interactive Query Audit

Table 2 documents representative key prompts submitted to the AI system across the project pipeline.

Table 2: Audit of Key Prompts and Functional Objectives

Phase	Core Functional Objective	Representative Prompt Snippet
Phase 1: Setup	Dataset download & Caroline filter implementation	<code>``Write a Python script using spaCy en_core_web_sm to replace PERSON, ORG, and GPE tags in FOLIO premises with synthetic Lewis Carroll tokens...``</code>
Phase 2: Inference	Multi-model zero-shot evaluation pipeline	<code>``Create a PyTorch loop evaluating Qwen2.5, Gemma-2, and Mistral sequentially with explicit memory cleanup via gc.collect() and empty_cache()...``</code>
Phase 3: Logic	Z3 SMT Proof by Contradiction	<code>``Implement a Python class using z3-solver to test if Premises and Not(Conclusion) is UNSAT for both original and Caroline text...``</code>
Phase 4: Surprisal	Pure PyTorch logit extraction	<code>``Bypass minicons and extract raw logits from final layers to compute conditional surprisal -log P(target context) using log_softmax and gather...``</code>
Phase 5: Writing	Academic paper formatting	<code>``Act as a Senior Academic Editor. Generate modular LaTeX sections formatted for Elsevier Expert Systems with Applications...``</code>

4 Human Oversight, Verification, and Rejection Audit

Per the professor’s rubric, AI-generated outputs were never accepted uncritically. Every code snippet was tested in Kaggle environments, and every textual statement was cross-verified against empirical outputs. Table 3 outlines major instances where AI-generated code or text contained flaws, hallucinations, or bugs, and details the corrective actions taken by the human author.

Table 3: Audit of AI Output Verification, Corrections, and Rejections

Target Area	AI Output / Error Detected	Human Author Corrective Action
Dataset Key Mapping	AI generated code using <code>df['example_id']</code> , causing a <code>KeyError</code> in Pandas.	Corrected the script to use fallback indexing (<code>df['id'] = df.index</code>), ensuring robust column selection.
H2 Synthetic Diversity	AI loop <code>for i in range(5)</code> created 50 minimal pairs with identical duplicate sentences.	Redesigned the data generator by introducing 5 distinct adjective modifiers (<i>massive, huge, gigantic, etc.</i>), producing 50 strictly unique pairs.
Multi-GPU Crash	AI suggested <code>minicons</code> library, which crashed with <code>AttributeError</code> under multi-GPU sharding.	Completely rejected the third-party wrapper; authored a pure PyTorch tensor function extracting raw logits directly via <code>log_softmax</code> .
Text Hallucination	AI generated unverified qualitative claims regarding Gemma’s and Mistral’s behavioral quirks.	Flagged and rejected the speculative text; rewrote the Discussion section to rely strictly on mathematically measured accuracy deltas (12.25%, 11.27%, 7.84%).
LaTeX Bad Boxes	AI output generated <code>Overfull \hbox</code> warnings due to long GitHub URLs.	Fixed by adding <code>\usepackage{xurl}</code> and modifying text wrappings to achieve 0 compilation errors.

5 Attestation of Academic Integrity

The author certifies that the experimental findings, vector plots, formal Z3 proofs, and statistical conclusions presented in the final manuscript represent an authentic, reproducible body of scientific work. All AI contributions have been transparently documented herein.

Student Signature: Alireza Ebrahimi

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